

```

import pandas as pd
import matplotlib.pyplot as plt
# Load dataset
df = pd.read_csv('Dataset_Day4.csv') # Replace with actual filename
# Select numeric columns only
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns
# --- Boxplot & Outlier Detection ---
for col in numeric_cols:
    plt.figure(figsize=(6, 4))
    plt.boxplot(df[col].dropna(), vert=True)
    plt.title(f'Boxplot for {col}')
    plt.ylabel(col)
    plt.grid(True)
    plt.show()

    # Outlier detection using IQR
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]
    if not outliers.empty:
        print(f"\nOutliers in '{col}':")
        print(outliers[[col]])
    else:
        print(f"No outliers in '{col}'\n")

```

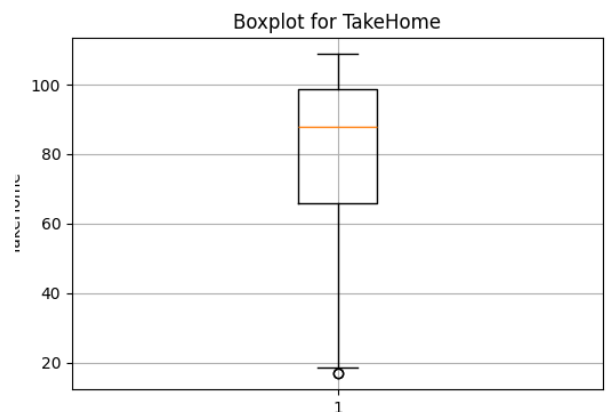
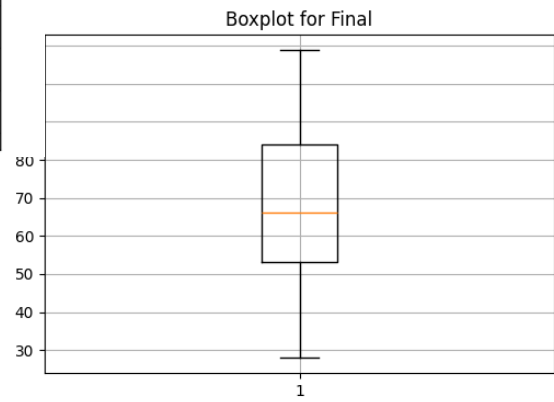
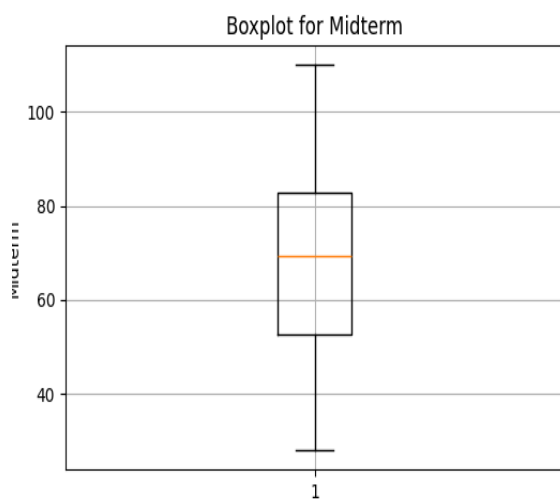
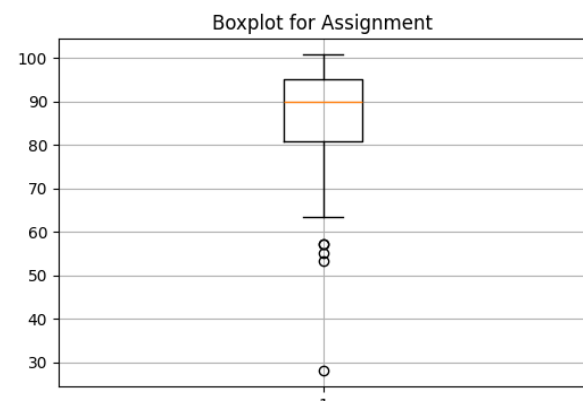
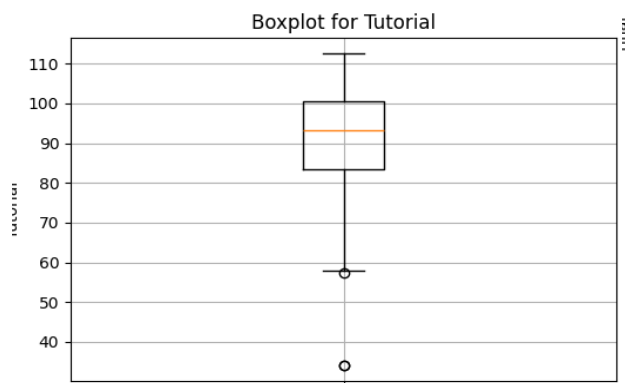
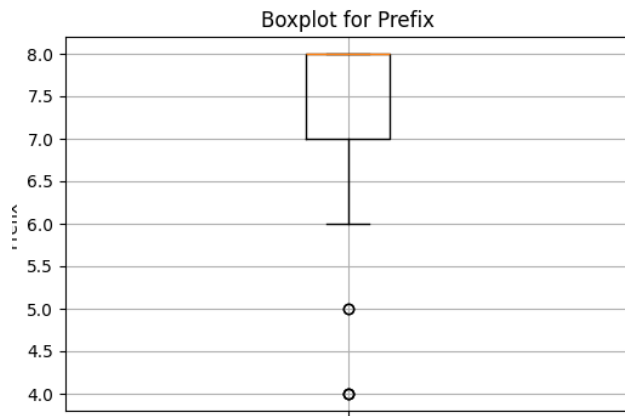
```

# --- Correlation Matrix ---
correlation_matrix = df[numeric_cols].corr()

print("\nCorrelation Matrix:")
print(correlation_matrix)

# Plot the correlation matrix using Matplotlib
plt.figure(figsize=(10, 8))
plt.imshow(correlation_matrix, cmap='coolwarm', interpolation='none')
plt.colorbar(label='Correlation Coefficient')
plt.xticks(range(len(numeric_cols)), numeric_cols, rotation=90)
plt.yticks(range(len(numeric_cols)), numeric_cols)
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()

```



```
(myenv) 101@a_04@101a:~/mlrc/d/python/day4$ py
```

```
Missing values before treatment:
```

```
Prefix      0
```

```
Assignment  0
```

```
Tutorial    0
```

```
Midterm     0
```

```
TakeHome    1
```

```
Final       3
```

```
dtype: int64
```

```
Missing values after treatment:
```

```
Prefix      0
```

```
Assignment  0
```

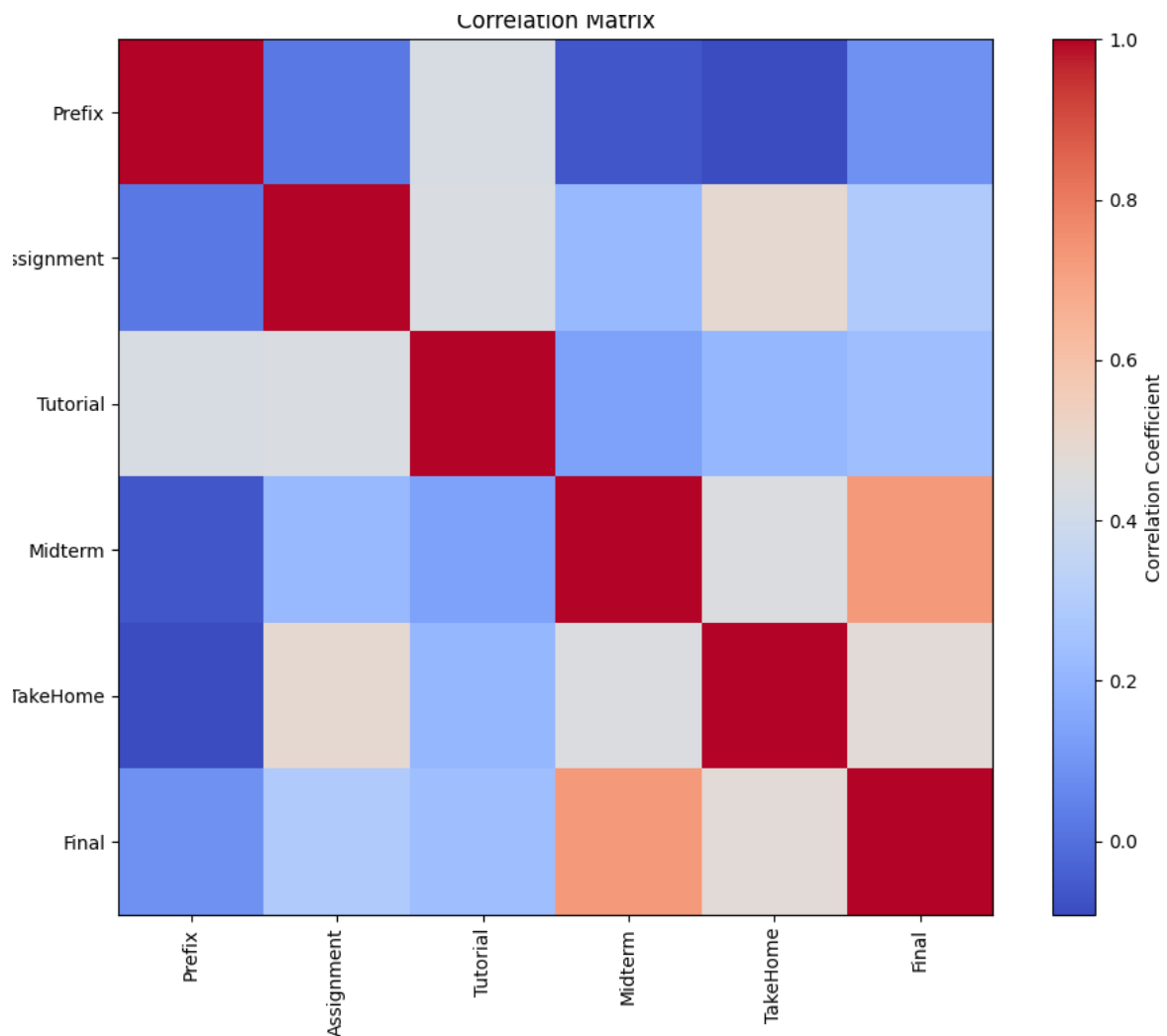
```
Tutorial    0
```

```
Midterm     0
```

```
TakeHome    0
```

```
Final       0
```

```
dtype: int64
```



```

1  import pandas as pd
2  df = pd.read_csv("Dataset_Day4.csv")
3
4  print("Missing values before treatment:")
5  print(df.isnull().sum())
6
7  # Fill missing numerical columns with mean or median
8  numerical_cols = df.select_dtypes(include=['float64', 'int64']).columns
9  for col in numerical_cols:
10 |     df[col] = df[col].fillna(df[col].mean())
11
12 # Fill missing categorical columns with mode
13 categorical_cols = df.select_dtypes(include=['object']).columns
14 for col in categorical_cols:
15 |     df[col] = df[col].fillna(df[col].mode()[0])
16
17 # Confirm all missing values are handled
18 print("\nMissing values after treatment:")
19 print(df.isnull().sum())

```